
Metastar: Predict tumor metabolic state from transcriptomic data

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Abstract

Cancer cells often shift between glycolysis and oxidative phosphorylation (OXPHOS), but metabolic state is not explicitly labeled in most gene expression datasets. In this work, we predict tumor metabolic phenotype using single-sample gene set enrichment analysis (ssGSEA) scores for glycolysis and OXPHOS and define a differential “delta” score to label tumors as Glycolytic, Oxidative, or Mixed. These labels are used to train an XGBoost classifier to predict metabolic state from gene expression data.

The key advantage is that this approach does not require any manually curated labels. The labels are generated automatically from the data itself. These results demonstrate that pathway-based labeling combined with machine learning provides a practical framework for studying tumor metabolism in large genomic datasets.

Code Availability

The implementation for this project is available at: <https://github.com/zackschwab/metastar>

1 Introduction

Cancer cells reprogram metabolism to support growth and survival. A well-known example is the Warburg effect, where tumors preferentially rely on glycolysis even in the presence of oxygen (1). However, some tumors may retain or revert to OXPHOS. This variability is important, as tumors with different metabolic programs can respond differently to therapies.

A key challenge is that metabolic state is rarely explicitly labeled in gene expression datasets. Instead, it must be inferred from high-dimensional molecular data. Existing approaches often rely on single-gene markers or curated annotations, which can be noisy, incomplete, or unavailable (2). Moreover, individual genes fail to capture the full depth of information across biological pathways.

To address this, we infer metabolic state at the pathway level. We compute ssGSEA scores for glycolysis and OXPHOS and define a differential “delta” score to capture their relative activity. These labels are used to train a supervised model that predicts metabolic type.

Contributions:

- A pathway-based method to infer tumor metabolic state using ssGSEA scores
- A delta-score labeling scheme to distinguish Glycolytic and Oxidative tumors
- A supervised model that predicts metabolic phenotype and validates inferred labels using known marker genes

31 2 Related Work

32 Prior work on cancer metabolism has shown that tumors differ in their reliance on glycolysis
33 and OXPHOS. Studies using glycolysis- or OXPHOS-related gene signatures have linked these
34 pathways to prognosis, survival, and immune response, suggesting that metabolic state is biologically
35 meaningful and clinically relevant (3; 7). However, many of these approaches focus on one pathway
36 at a time.

37 Machine learning has been widely applied to cancer classification using gene expression data. These
38 methods can learn patterns from high-dimensional transcriptomic features and have been used for
39 tasks such as tumor subtype classification, prognosis prediction, and biomarker discovery.

40 Pathway-based methods such as GSEA and ssGSEA provide a way to summarize gene expression into
41 interpretable biological activity scores. Rather than treating thousands of genes independently, these
42 methods estimate pathway activity at the sample level, making them useful for studying biological
43 processes that involve coordinated gene regulation (4).

44 Our work combines these directions. We infer metabolic labels from pathway-level glycolysis and
45 OXPHOS activity, then train a supervised classifier to predict those labels from gene expression data.
46 This differs from prior gene-signature approaches by explicitly modeling the balance between two
47 metabolic programs and differs from standard supervised cancer classification by addressing a setting
48 where the target label must first be inferred.

49 3 Methodology

50 3.1 Problem formulation

51 We formulate tumor metabolic state prediction as a supervised classification problem over gene
52 expression profiles. Let $X \in \mathbb{R}^{n \times p}$ represent the expression matrix, where n is the number of tumor
53 samples and p is the number of genes. Each sample x_i is represented by its TPM-normalized gene
54 expression values. The goal is to learn a function

$$f(x_i) \rightarrow y_i$$

55 where y_i represents the inferred metabolic phenotype of sample i .

56 Because metabolic phenotype is not directly provided, we first infer labels using pathway-level
57 activity. We compute ssGSEA scores for glycolysis and OXPHOS and define a metabolic balance
58 score:

$$\Delta_i = S_{\text{glycolysis},i} - S_{\text{OXPHOS},i}$$

59 where each term represents the normalized pathway enrichment score for sample i .

60 Samples are labeled using fixed thresholds:

$$y_i = \begin{cases} \text{Glycolytic,} & \Delta_i > 0.5 \\ \text{Oxidative,} & \Delta_i < -0.5 \\ \text{Mixed,} & -0.5 \leq \Delta_i \leq 0.5 \end{cases}$$

61 For the final machine learning task, Mixed samples are excluded to create a binary classification
62 problem.

63 3.2 Expression matrix construction

64 RNA-seq expression data from TCGA-KIRC samples were processed to construct a gene-by-sample
65 matrix using TPM-normalized values (10; 11). Gene identifiers were standardized and duplicate
66 entries were removed to ensure consistency across samples.

67 The resulting matrix was used both for pathway scoring via ssGSEA and as the feature matrix for
68 supervised model training.

69 3.3 Pathway-based metabolic labeling

70 To infer metabolic phenotype, we used ssGSEA to compute sample-level enrichment scores for
71 predefined hallmark pathways (4). Specifically, we focused on HALLMARK_GLYCOLYSIS and
72 HALLMARK_OXIDATIVE_PHOSPHORYLATION, which represent two major modes of tumor energy
73 metabolism (8; 9).

74 The resulting enrichment scores were standardized across samples using z-score normalization. The
75 delta score Δ captures the relative dominance of glycolysis versus OXPHOS, with positive values
76 indicating glycolytic bias and negative values indicating oxidative bias.

77 This approach reflects pathway-level activity rather than relying on individual genes.

78 3.4 Machine learning dataset preparation

79 After assigning metabolic labels, we constructed a supervised dataset by pairing each sample’s gene
80 expression vector with its inferred phenotype. Samples labeled as Mixed were excluded to reduce
81 ambiguity and focus the model on distinguishing Glycolytic and Oxidative states.

82 The final dataset consists of gene expression features and a binary metabolic label.

83 3.5 Model training

84 We trained an XGBoost classifier to predict metabolic phenotype from gene expression data (5). XG-
85 Boost was chosen for its ability to model nonlinear relationships and interactions in high-dimensional
86 data.

87 The model is trained on gene expression features to learn patterns that distinguish Glycolytic and
88 Oxidative tumors.

89 3.6 Biological validation

90 To assess whether inferred labels capture meaningful biology, we evaluated their association with
91 known metabolic marker genes. In particular, we examined *HK2*, a glycolysis-associated gene
92 expected to be elevated in Glycolytic tumors.

93 We compared gene expression between metabolic groups and computed correlations between pathway
94 scores and selected genes to verify consistency between inferred labels and known metabolic behavior.

95 4 Experiments

96 4.1 Dataset and preprocessing

97 We evaluated our approach using RNA-seq gene expression data from The Cancer Genome Atlas
98 Kidney Renal Clear Cell Carcinoma (TCGA-KIRC) cohort (3). The dataset consists of tumor samples
99 with transcript-level expression quantified in TPM (transcripts per million). After preprocessing and
100 filtering, each sample is represented by thousands of gene expression features.

101 Raw data was loaded from per-sample TSV files and processed to construct a consistent expression
102 matrix. Preprocessing steps included removing non-gene entries, filtering invalid or missing gene
103 identifiers, and deduplicating genes by name. Expression values were indexed by gene and combined
104 into a gene-by-sample matrix.

105 To generate labels, we computed ssGSEA enrichment scores for the HALLMARK_GLYCOLYSIS and
106 HALLMARK_OXIDATIVE_PHOSPHORYLATION pathways. These scores were standardized across sam-
107 ples using z-score normalization. A delta score was then computed as the difference between
108 glycolysis and OXPHOS scores, and samples were labeled as Glycolytic, Oxidative, or Mixed based
109 on fixed thresholds.

110 For machine learning, samples labeled as Mixed were excluded, resulting in a binary classification
111 dataset consisting of Glycolytic and Oxidative samples. The final dataset was summarized in terms
112 of total samples, feature dimensionality, and class distribution to ensure suitability for training.

113 4.2 Train/test split

114 We partitioned the dataset into training and testing sets using a stratified 80/20 split. All model
115 selection and hyperparameter tunings were performed on the training set, while the test set was held
116 out for final evaluation.

117 4.3 Model training setup

118 We trained an XGBoost classifier to predict metabolic phenotype from gene expression data. Hyper-
119 parameter tuning was performed using randomized search with 5-fold cross-validation on the training
120 set. The search space included key parameters such as number of trees, maximum depth, learning
121 rate, subsampling ratio, feature sampling ratio, and regularization strength.

122 The best performing hyperparameter configuration was selected based on mean cross-validated
123 ROC-AUC. The final model was then trained on the full training set using these optimized parameters
124 and evaluated on the held-out test set.

125 4.4 Evaluation metrics

126 Model performance was evaluated using several standard classification metrics:

- 127 • **ROC-AUC:** Measures the model’s ability to distinguish between Glycolytic and Oxidative
128 samples across all classification thresholds.
- 129 • **Accuracy:** The proportion of correctly classified samples.
- 130 • **Precision, Recall, and F1-score:** Provide additional insight into class-specific performance
131 and the trade-off between false positives and false negatives.

132 In addition to classification metrics, we also computed Spearman correlation coefficients between
133 pathway scores (glycolysis, OXPHOS, and delta) and selected gene expression levels (e.g., *HK2*,
134 *UQCRC1*, *RUNX1*). These correlations provide additional biological validation by assessing the
135 relationship between pathway activity and expression of metabolically relevant genes.

136 4.5 Reproducibility

137 All experiments were conducted using a fixed random seed to ensure reproducibility of data splits and
138 model training. The pipeline is fully deterministic given the input data and configuration, including
139 data preprocessing, ssGSEA scoring, label generation, dataset construction, and model training. The
140 use of standardized preprocessing steps and explicit hyperparameter search ranges ensures that results
141 can be reproduced and extended in future work.

142 5 Results

143 5.1 Metabolic Label Distribution

144 Applying ssGSEA and thresholding produces a metabolically heterogeneous cohort (Table 1). Approx-
145 imately 37% of tumors are Mixed, indicating that a substantial fraction of KIRC retains intermediate
146 metabolic activity and would be poorly served by a forced binary classification.

Table 1: Inferred Metabolic Label Distribution ($n = 614$)

Label	Count	Percentage
Mixed	227	37.0%
Glycolytic	226	36.8%
Oxidative	161	26.2%

147 The delta score distribution is roughly unimodal and centered near zero, with shoulders extending
148 into the Glycolytic and Oxidative tails, confirming that the 0.5 threshold selects for the extremes of a
149 continuous spectrum rather than imposing an artificial dichotomy.

5.2 Biological Validation of Inferred Labels

Spearman correlations between pathway scores and selected genes reveal strong, directionally consistent relationships (Table 2). *HK2* correlates positively with glycolysis as seen in Figure 1 ($\rho = +0.534$) and negatively with OXPHOS ($\rho = -0.429$), while *UQCRC1* shows the inverse pattern, strongly correlating with OXPHOS ($\rho = +0.651$). *RUNX1* exhibits the strongest association, with a large negative correlation to OXPHOS ($\rho = -0.731$) and a strong positive correlation to delta ($\rho = +0.802$).

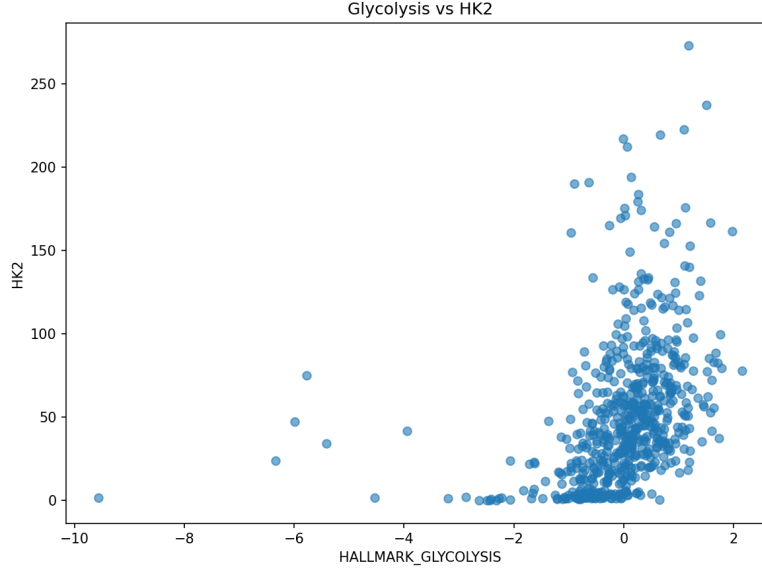


Figure 1: **Relationship between glycolysis enrichment and HK2 expression.** Scatterplot showing the positive association between ssGSEA glycolysis enrichment scores and *HK2* expression across TCGA-KIRC tumor samples. Higher glycolytic activity corresponds to increased *HK2* expression, supporting the biological validity of the inferred metabolic labels.

Table 2: Spearman Correlations Between Pathway Scores and Gene Expression

Comparison	ρ
HALLMARK_GLYCOLYSIS vs <i>HK2</i>	+0.534
HALLMARK_OXIDATIVE_PHOSPHORYLATION vs <i>HK2</i>	-0.429
Δ vs <i>HK2</i>	+0.619
HALLMARK_GLYCOLYSIS vs <i>UQCRC1</i>	-0.059
HALLMARK_OXIDATIVE_PHOSPHORYLATION vs <i>UQCRC1</i>	+0.651
Δ vs <i>UQCRC1</i>	-0.452
HALLMARK_GLYCOLYSIS vs <i>RUNX1</i>	+0.460
HALLMARK_OXIDATIVE_PHOSPHORYLATION vs <i>RUNX1</i>	-0.731
Δ vs <i>RUNX1</i>	+0.802

The Mann-Whitney test for *HK2* confirms that Glycolytic tumors significantly overexpress the gene (Table 3). The 11.2-fold median increase and minuscule p -value (1.0×10^{-45}) provide strong evidence that the ssGSEA-derived labels capture genuine metabolic divergence rather than random transcriptional noise.

5.3 Machine Learning Performance

The binary classification dataset contains 387 samples and 59,427 features (Table 4).

Hyperparameter search over 50 randomized configurations yields consistently high and stable cross-validated performance. The best configuration (Table 5) achieves a mean ROC-AUC of 0.9981, with

Table 3: HK2 Biological Validation

Metric	Value
Glycolytic median TPM	62.38
Oxidative median TPM	5.58
Fold change	11.2×
Mann-Whitney U p -value	1.0×10^{-45}
Significant ($\alpha = 0.05$)	Yes

Table 4: ML Dataset Summary

Property	Value
Total samples	387
Features (genes)	59,427
Glycolytic (1)	226 (58.4%)
Oxidative (0)	161 (41.6%)

individual fold scores ranging from 0.9968 to 0.9989. The mean training score for this configuration is 1.0000, yielding a small generalization gap of 0.0019, which suggests minimal overfitting despite the high dimensionality.

Table 5: Top Hyperparameter Search Results (5-Fold CV)

Rank	Mean ROC-AUC	Best Fold	Key Parameters
1	0.9981	0.9989	subsample=0.5, n_estimators=300, learning_rate=0.1 reg_alpha=0.1, max_depth=6,
2	0.9970	0.9989	subsample=0.5, n_estimators=100, learning_rate=0.2 reg_alpha=0.1, max_depth=3,
3	0.9964	0.9989	subsample=0.7, n_estimators=800, learning_rate=0.1 reg_alpha=0.001, max_depth=5,
4	0.9951	0.9989	subsample=1.0, n_estimators=1200, learning_rate=0.3 reg_alpha=0.001, max_depth=2,
5	0.9946	0.9989	subsample=0.5, n_estimators=100, learning_rate=0.2 reg_alpha=0.001, max_depth=5,

On the held-out test set, the final tuned XGBoost model achieved a ROC-AUC of 0.985, demonstrating strong discriminative ability (Table 6). Balanced precision and recall across both classes further indicate robust classification performance despite class imbalance.

The confusion matrix implied by per-class recalls (94% for Oxidative, 93% for Glycolytic) shows only 2 misclassified Oxidative samples and 3 misclassified Glycolytic samples out of 78. This level of performance suggests that the transcriptomic signatures separating Glycolytic and Oxidative KIRC are strong and coherent.

6 Discussion

Our results support the main idea behind Metastar: pathway-level inference can reliably predict tumor metabolic state using standard RNA-seq data. The model achieved a high test ROC-AUC of 0.985, showing that Glycolytic and Oxidative phenotypes can be distinguished accurately. Performance also remained strong between cross-validation and the held-out test set, with only a small drop from a mean CV AUROC of 0.998 to a test AUROC of 0.985. In addition, the small train-test gap

Table 6: Final Model Performance on Held-Out Test Set ($n = 78$)

Metric	XGBoost (Tuned)
Accuracy	0.936
ROC-AUC	0.985
Precision (Oxidative)	0.910
Recall (Oxidative)	0.940
Precision (Glycolytic)	0.960
Recall (Glycolytic)	0.930
Macro F1	0.930

(0.0019) suggests that the regularization settings, including `reg_alpha` and `subsample`, helped limit overfitting despite the high-dimensional nature of the dataset.

Biologically, the validation analysis supports the reliability of the ssGSEA labels. Glycolytic tumors showed an 11.2-fold increase in *HK2* expression, which is consistent with the Warburg effect, where hexokinase-2 controls the first major step of glucose metabolism (6). In addition, the strong positive correlation between Δ and *RUNX1* ($\rho = +0.802$) suggests that *RUNX1* may play an important role in glycolytic metabolic reprogramming in KIRC. On the other hand, the strong association between *UQCRC1* and the OXPHOS score ($\rho = +0.651$) indicates that the oxidative subtype reflects active mitochondrial respiration rather than simply the absence of glycolysis.

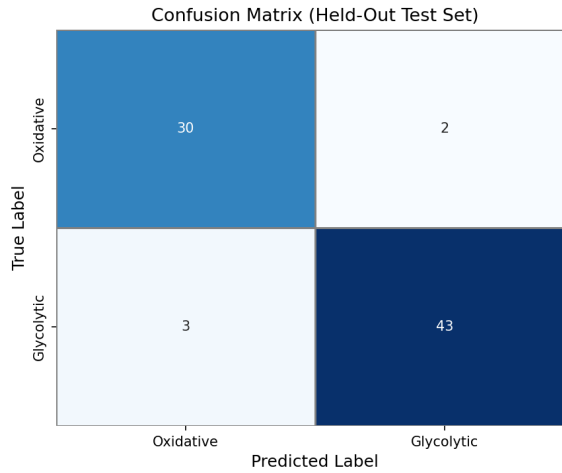


Figure 2: **Confusion matrix on the held-out test set** ($n = 78$). The model correctly classifies 30 of 32 Oxidative samples and 43 of 46 Glycolytic samples, yielding an overall accuracy of 93.6%. Only 5 samples are misclassified in total, confirming strong discriminative performance across both metabolic states.

The identification of 227 Mixed tumors (37%) highlights that tumor metabolism is better represented as a spectrum than as a strict binary state. Although Mixed samples were excluded from classification to improve label quality, future studies could model metabolism as a continuous prediction task using Δ or as a three-class classification problem. This approach may better capture intermediate metabolic states with distinct therapeutic responses.

XGBoost achieved strong predictive performance, showing the importance of modeling non-linear relationships between genes. This result is biologically reasonable because metabolic regulation depends on complex interactions such as feedback loops, enzyme saturation, and pathway crosstalk.

7 Limitations

Several limitations should be acknowledged. First, the study is restricted to a single cancer type (TCGA-KIRC). Renal cell carcinoma is known to be highly vascular and often reliant on fatty acid oxidation, so the exact thresholds and possibly the classifier itself may not generalize to other tissues without retraining or domain adaptation (3).

Second, the ± 0.5 threshold on Δ is heuristic. While it selects for metabolic extremes, different thresholds would alter class balance and could change model performance. A sensitivity analysis on threshold choice is recommended but was beyond the scope of this project.

Third, by excluding Mixed tumors, the study removes over one-third of the cohort from classifier training. In a clinical setting, these intermediate tumors may be the most biologically relevant cases, yet they are also the most difficult to classify reliably. A continuous scoring framework or probabilistic classification approach may better capture this metabolic spectrum.

Finally, transcriptomic data alone provide only a partial representation of tumor metabolism. While gene expression profiles capture transcriptional activity, they do not directly reflect protein abundance, post-translational modification, metabolite availability, or microenvironmental influences that shape the metabolic picture (12).

8 Conclusion

We introduce *Metastar*, an end-to-end framework that predicts tumor metabolic phenotype from bulk gene expression. The method uses ssGSEA pathway scores as input features and applies an XGBoost classifier to distinguish metabolic states. Applied to 614 TCGA-KIRC samples, the model identifies biologically meaningful Glycolytic and Oxidative subtypes, shows strong agreement with the glycolysis marker *HK2*, and achieves a test ROC-AUC of 0.985. Overall, these results show that tumor metabolic state can be accurately inferred from RNA-seq data using pathway-informed machine learning.

Future work will extend this framework to pan-cancer datasets to learn both shared and cancer-specific metabolic patterns. We also plan to integrate additional omics data, such as proteomics and metabolomics, to improve metabolic prediction. Ultimately, *Metastar* could serve as a fast and cost-effective tool for identifying patients who may benefit from metabolism targeted therapies using standard RNA-seq data.

9 Usage of AI Tools

AI tools were used to help revise grammar, improve clarity, and organize the report structure. AI was also used to help debug code, provide code suggestions, and format print statements.

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